

Diego Domenzain

PhD in Computational Geophysics

personal

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about me

I am a Mathematician with keen interest on applied problems and physics. Currently, I design better ways to explore the interior of the Earth at shallow depths. I use numerical modeling of partial differential equations, mathematical non-linear optimization, signal processing (in time and frequency), and high performance computing.

The non-linear optimization algorithms I have designed, implemented, and developed have been benchmarked as faster, lighter on memory, and of higher resolution than current industry standards.

Public proof of this statement can be found in the peer-review publications I have authored as a scientific researcher.

professional experience

Computational Geophysicist, R&D

2023 – today

Seequent - Labs, DK

Design and implementation of modeling and inversion of 1D electromagnetic diffusive field propagation for GPU accelerated compute that is 10x to 100x faster than its CPU implementation. Design and implementation of a 3D measuring protocol scheme for DC data measured in arbitrary spatial configurations. Graphics and interactive display were built in collaboration with a Canadian work colleague for an internal Hackathon event. Collaboration with the Product Management department for scraping and data-mining large amounts of geophysics mining reports. Due to lack of resources and urgency of the project, the AI processing was successfully developed to run on a local PC and without the use of API keys.

Computational Geophysicist

2023 – 2024

Seequent - AGS, DK

Design and implementation of a 3D visualization scheme for DC data measured in arbitrary spatial configurations. Currently used commercially in the 2D plotting engine of ResXInv. Implementation of an external consultant's machine learning algorithm for time-domain EM data filtering, ported from pseudo code to C++/ONNX for seamless integration in the Delphi-based user interface. Development of the Parquet-Arrow read/write interface for the AGS desktop products. Thorough review of the physics and source code of the legacy modeling engine in the desktop product Workbench.

Postdoctoral Researcher

2021 – 2023

Aarhus Universitet, DK

Design and implementation of a 3D DC inversion algorithm capable of handling two orders of magnitude more data than previous schemes. Worked closely with the design of a novel survey instrument to achieve the needs of our industry partner. The project specifically aimed (and was successful) at delineating the 3D spread of a contaminant-remediation agent in the subsurface.

Postdoctoral Researcher

2020 – 2021

Colorado School of Mines, USA

Design and implementation of a multi-physics joint inversion of electromagnetic waves (radar) and DC data acquired at the surface. The result of this work was the first successful surface-acquired full-waveform inversion of radar data. Also developed a machine learning method for enhancing resolution of recovered parameters using remote sensing data.

PhD Researcher

2015 – 2020

Boise State University, USA

Joined an existing NSF funded project aiming to join DC voltages and electromagnetic radar data in order to recover 2D subsurface electrical parameters. Successfully designed and developed such a scheme with the added bonus of a new electrical resistivity inversion algorithm capable of handling large amounts of data using low memory requirements. Moreover, through the use of numerical curvature algorithms it was possible to enhance resolution in areas of nearly zero sensitivity.

skills

<i>Languages</i>	Fortran, C/C++, Cuda, OpenCL, Julia, Python, Matlab, Bash, PowerShell, Latex, PyTorch, TensorFlow.
<i>Software ethos</i>	Low level implementation of non-linear optimization methods and physics numerical modeling. Design of parallel computing code with memory, speed, and accuracy performance in mind. Implementation of high performance computing in the cloud.

publicly available code

alles

since 2020

Personal dossier of projects not directly affiliated with any of my previous positions. Topics range from graph theory algorithms, partial differential equations, mathematical optimization, and scientific visualization of data. Written in C, Fortran, C++, Cuda, OpenCL, Shell, Python, and Matlab. Hosted in Github.

Gerjoi

since 2015

Exploration geophysics software. Forward modeling of Poisson's equation and the full electromagnetic wave equation (radar) for non-homogeneous 2D media. Non-linear multi-physics inversion of measured electric fields and voltages. Uses the full-waveform inversion (FWI) algorithm in the case of radar. Up to date this is the only software capable of inverting surface acquired radar data using the FWI scheme. Written in Matlab. Hosted in Github.

references

Available upon request

peer reviewed publications

3D DC inversion, visualization, and processing of dense time-lapse data in fine domains applied to remediation monitoring

2023

Diego Domenzain, Lichao Liu, Iván Yélamos Vela, and Anders Vest. Geophysics

Joint full-waveform ground-penetrating radar and electrical resistivity inversion applied to field data acquired on the surface

2022

Diego Domenzain, John Bradford, and Jodi Mead. Geophysics

Efficient inversion of 2.5D electrical resistivity data using the discrete adjoint method

2021

Diego Domenzain, John Bradford, and Jodi Mead. Geophysics

Joint inversion of full-waveform inversion GPR and ER data. Part 2

2020

Diego Domenzain, John Bradford, and Jodi Mead. Geophysics

Joint inversion of full-waveform inversion GPR and ER data. Part 1

2020

Diego Domenzain, John Bradford, and Jodi Mead. Geophysics

Maximal arcs, above and beyond

2014

Michigan Technological University, Master Thesis

Construction and dissection of finite geometries using algebraic and combinatorial methods.

Surface Error Correcting Codes

2012

UNAM, Bachelor Thesis

Construction of quantum error correction codes using group theory, algebraic topology, and graph theory.